

Astronomy at University of Groningen

Your interests: **Space exploration and astronomy**

Official programme ↗

Astronomy and astrophysics

Introduction Astronomy
Physics of Stars
Physics of Galaxies
Radiative Processes in Astrophysics
Advanced Electrodynamics
Astrophysical Hydrodynamics
Interstellar Medium

Physics and mathematics

Mechanics and Relativity — 10 EC
Electricity and Magnetism — 10 EC
Thermal Physics — 10 EC
Quantum Physics 1
Quantum Physics 2
Waves and Optics
Calculus 1 (for Physics)
Calculus 2 (for Physics)
Linear Algebra (for Physics)
Mathematical Physics
Complex Analysis

Computing, data and observation

Introduction to Programming and Computational Methods
Numerical Methods
Statistics for Astronomy
Observational Astronomy

Exploration, instruments and responsible technology

Minor: Instrumentation & Informatics — 30 EC

Earth, planets and life

Introduction to Planetary Science

Research and academic development

Physics Lab: Skills
Astronomy Bachelor Research Project — 15 EC

Closest match to Astronomy

Your interests: **Space exploration and astronomy**

UCR courses ↗

Physics and mathematics

- Advanced Calculus · 7.5 EC
- Linear Algebra · 7.5 EC
- Mathematical Methods · 7.5 EC
- Logic & Discrete Mathematics · 7.5 EC
- Linear Systems · 7.5 EC

Computing, data and observation

- Introduction to Programming · 7.5 EC
- Computational & Numerical Methods · 7.5 EC
- Probability & Statistics · 7.5 EC
- Earth Observation · 7.5 EC
- Introduction to Data Science · 7.5 EC
- Machine Learning · 7.5 EC
- Image Processing & Computer Vision · 7.5 EC
- Neural Networks & Deep Learning · 7.5 EC
- Optimization Methods in AI · 7.5 EC
- Reinforcement Learning · 7.5 EC
- Algorithms & Data Structures · 7.5 EC
- Networks & Operating Systems · 7.5 EC
- Database Management Systems · 7.5 EC
- Earth System Science Tools · 7.5 EC

Exploration, instruments and responsible technology

- Robotics · 7.5 EC
- Consumer Product Design · 7.5 EC

Earth, planets and life

- Sustainability Essentials I · 7.5 EC
- Environmental Biogeochemistry · 7.5 EC

Research and academic development

- Personal & Professional Development · 7.5 EC

UCR does not offer astronomy or physics courses; this pathway builds the closest feasible depth through mathematics, computing, data and observation.

Astronomy + related subjects

Your interests: **Space exploration and astronomy**

UCR courses ↗

Physics and mathematics

Linear Algebra · 7.5 EC

Mathematical Methods · 7.5 EC

Logic & Discrete Mathematics · 7.5 EC

Computing, data and observation

Introduction to Programming · 7.5 EC

Computational & Numerical Methods · 7.5 EC

Probability & Statistics · 7.5 EC

Earth Observation · 7.5 EC

Introduction to Data Science · 7.5 EC

Machine Learning · 7.5 EC

Image Processing & Computer Vision · 7.5 EC

Optimization Methods in AI · 7.5 EC

Earth System Science Tools · 7.5 EC

Exploration, instruments and responsible technology

Robotics · 7.5 EC

Consumer Product Design · 7.5 EC

Renewable System Design · 7.5 EC

AI Ethics · 7.5 EC

Earth, planets and life

Sustainability Essentials I · 7.5 EC

Sustainability Essentials II · 7.5 EC

Dynamic Earth · 7.5 EC

Environmental Biogeochemistry · 7.5 EC

Earth's Resources · 7.5 EC

Climate Change · 7.5 EC

Research and academic development

Research in a Sustainable Delta · 7.5 EC

Personal & Professional Development · 7.5 EC

A broader programme around your interests

Your interests: **Space exploration and astronomy**

UCR courses ↗

Physics and mathematics

Mathematical Methods · 7.5 EC

Computing, data and observation

Introduction to Programming · 7.5 EC

Earth Observation · 7.5 EC

Introduction to Data Science · 7.5 EC

Machine Learning · 7.5 EC

Image Processing & Computer Vision · 7.5 EC

Neural Networks & Deep Learning · 7.5 EC

Earth System Science Tools · 7.5 EC

Exploration, instruments and responsible technology

Robotics · 7.5 EC

Consumer Product Design · 7.5 EC

Renewable System Design · 7.5 EC

AI Ethics · 7.5 EC

Earth, planets and life

Sustainability Essentials I · 7.5 EC

Sustainability Essentials II · 7.5 EC

Dynamic Earth · 7.5 EC

Environmental Biogeochemistry · 7.5 EC

Earth's Resources · 7.5 EC

Climate Change · 7.5 EC

Introduction to Biodiversity · 7.5 EC

The Science of Evolution · 7.5 EC

Foundation of Life, Health & Biomedicine · 7.5 EC

Biochemistry · 7.5 EC

Research and academic development

Research in a Sustainable Delta · 7.5 EC

Personal & Professional Development · 7.5 EC

UCR does not offer astronomy or physics courses; this pathway broadens into Earth systems, life science, data and exploration technology.